Solution

Matrix Theory, Quiz 8S — April 17, 2026

Given $A = \begin{bmatrix} 2 & 1 \\ 2 & 2 \\ 3 & -2 \end{bmatrix}$ and $\mathbf{z} = \begin{bmatrix} 35 \\ 6 \\ 1 \end{bmatrix}$, let W denote the column space of A.

(1) Write the vector $\mathbf{z}$ in the form $\mathbf{z} = \text{proj}_W(\mathbf{z}) + \text{orth}_W(\mathbf{z})$, where $\text{proj}_W(\mathbf{z})$ is the orthogonal projection of $\mathbf{z}$ onto W and $\text{orth}_W(\mathbf{z})$ is a vector orthogonal to W.

(2) Find the distance from $\mathbf{z}$ to W.

(3) True or False? The equation $A\mathbf{x} = \mathbf{z}$ has a solution.

(4) True or False? The system of linear equations: $\begin{cases} 2x + y = 35 \\ 2x + 2y = 6 \\ 3x - 2y = 1 \end{cases}$ is consistent.

$$\begin{array}{r} 17 \\ 5 \\ \hline 85 \end{array}$$

1) $D = A^T A = \begin{bmatrix} 17 & 0 \\ 0 & 9 \end{bmatrix}$

$\text{proj}(z) = A D^{-1} A^T \begin{bmatrix} 35 \\ 6 \\ 1 \end{bmatrix} = A D^{-1} \begin{bmatrix} 2\cdot 35 + 2\cdot 6 + 3 \\ 35 + 2\cdot 6 - 2 \end{bmatrix} = A \begin{bmatrix} 85/17 \\ 45/9 \end{bmatrix} = A \begin{bmatrix} 5 \\ 5 \end{bmatrix}$

$\text{proj}(z) = 5\begin{bmatrix} 2 \\ 2 \\ 3 \end{bmatrix} + 5\begin{bmatrix} 1 \\ 2 \\ -2 \end{bmatrix} = \begin{bmatrix} 15 \\ 20 \\ 5 \end{bmatrix} = 5\begin{bmatrix} 3 \\ 4 \\ 1 \end{bmatrix}$

1 $\text{orth}(z) = z - \text{proj}\, z = \begin{bmatrix} 20 \\ -14 \\ -4 \end{bmatrix}$

$40 - 28 - 12 = 0$

1 (2) $\|\text{orth}\, z\| = \sqrt{(20)^2 + (14)^2 + (4)^2}$

1 (3) False

1 (4) False

$D = \begin{bmatrix} 9 & 0 \\ 0 & 17 \end{bmatrix}$

Version 8R : (1)

$\text{proj}(z) = A D^{-1} A^T \begin{bmatrix} 35 \\ 6 \\ 1 \end{bmatrix} = A \begin{bmatrix} 1/9 & 0 \\ 0 & 1/17 \end{bmatrix} \begin{bmatrix} 45 \\ 85 \end{bmatrix}$

$= A \begin{bmatrix} 5 \\ 5 \end{bmatrix} = \begin{pmatrix} 15 \\ 20 \\ 5 \end{pmatrix} = 5\begin{pmatrix} 3 \\ 4 \\ 1 \end{pmatrix}$

$\text{orth}(z) = z - \text{proj}\, z = \begin{bmatrix} 20 \\ -14 \\ -4 \end{bmatrix}$

(2) $\|\text{orth}\, z\| = \sqrt{(20)^2 + (14)^2 + (4)^2}$

(3) False (4) False